Supply and Demand

Buyers and sellers → curves and schedules → equilibrium → market changes

1. The law of demand

Begin with a market

A market brings together people who want to buy something and people willing to sell it. We will use the *market for bicycle-repair appointments in one town*: how many appointments customers want each week and how many local workshops are willing to provide. The price is dollars per appointment; the quantity is appointments per week. Keeping both units explicit will make the tables and diagrams easier to read.

> **DEFINITION Market and competitive market**
>
> A **market** is a group of buyers and sellers of a particular good or service. In a **competitive market**, many buyers and sellers take the market price as given; no single participant can substantially change it. A **perfectly competitive market** also has an essentially identical product across sellers. Supply-and-demand graphs begin with this useful benchmark.

An individual workshop might prefer to charge more, but customers can turn to its competitors. A customer might prefer a lower price, but cannot dictate it to the entire market. The interaction of all buyers and sellers helps determine the price. The model is a simplifying approximation, not a claim that every real market is perfectly competitive.

> **DEFINITION Quantity demanded, demand schedule, and demand curve**
>
> **Quantity demanded** is the amount buyers are willing *and able* to buy at a specified price during a specified period. A **demand schedule** lists quantities demanded at different prices. A **demand curve** plots those price–quantity pairs, with price on the *vertical* axis and quantity on the *horizontal* axis.

> **KEY RESULT The law of demand — required principle 1**
>
> **Other things equal**, a higher price causes a lower quantity demanded; a lower price causes a higher quantity demanded. This is a *movement along* an unchanged demand curve, not a shift of the curve.

Why does demand slope downward?

Consider someone deciding whether to arrange an additional bike repair. At a low price, it may be worth fixing a minor problem; at a higher price, the same person may postpone the job or do it at home. Some customers with less urgent repairs stop buying sooner than others. Thus, as price rises, fewer appointments are worthwhile. This is a statement about buyers' *choices at different prices*, with their incomes, preferences, and other prices held fixed.

Worked example: Read a demand schedule

The numbers below are a hypothetical *market* demand schedule. At $25, customers want 70 appointments per week; at $40, they want 40. The extra $15 per appointment reduces quantity demanded by 30 appointments. In this example the law of demand is visible directly in the numbers.

Price ($)	20	25	30	35	40	45	50
Quantity demanded (weekly)	80	70	60	50	40	30	20

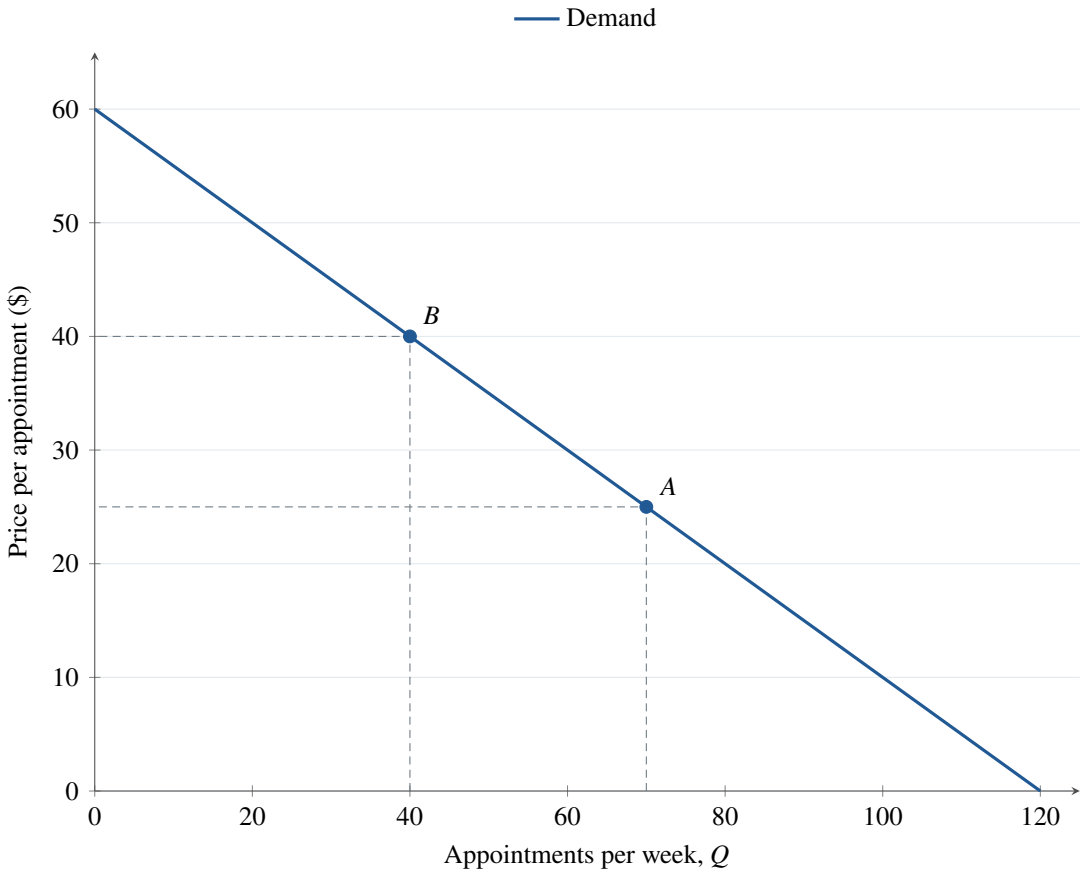


Figure 1. The law of demand. Moving from *A* to *B* means a higher price and a smaller quantity demanded *on the same curve*.

Read the graph. Find $25 on the vertical axis; move right to demand and down to 70 appointments. At $40 the same procedure gives 40. The arrow is not necessary: the two points show the direction of the relationship. We have not changed any factor other than price.

How individual demands become market demand

There is an important distinction between *one buyer's* demand and *market* demand. A single rider may be willing to pay $45 for an urgent repair but only $20 for a less urgent second repair. That rider's quantity demanded rises when the price falls. To find total market demand, add up what *every* rider wants at the *same price*.

KEY RESULT Market demand is horizontal addition

At each price, **market quantity demanded = sum of the quantities demanded by all buyers**. We add quantities horizontally because they are measured on the horizontal axis. We do *not* add prices or individual willingness to pay at a given quantity.

Worked example: Add two groups of customers

For illustration, suppose riders from the north and south parts of town are the only two groups buying repairs. At \$30, north-side riders want 36 appointments and south-side riders want 24; market demand is 36 + 24 = 60. At \$40, they want 24 and 16; market demand is 40. The aggregation rule is the same whether there are two customers or two thousand.

Price per appointment	North-side customers	South-side customers	Market demand
\$20	48	32	80
\$30	36	24	60
\$35	30	20	50
\$40	24	16	40
\$50	12	8	20

Each column answers a different question. The first two tell us who wants the service. The last tells us how many appointments the market demands. Only the last is the *market* demand curve.

What shifts the demand curve?

The law of demand compares different prices while other influences stay fixed. In reality those influences can change. When buyers want more appointments at *every given price*, the entire demand curve shifts *right*. When they want fewer at every price, it shifts *left*. That is a change in *demand*, not merely a change in *quantity demanded*.

DEFINITION Normal and inferior goods; substitutes and complements

For a **normal good**, higher income increases demand, other things equal. For an **inferior good**, higher income reduces demand. These labels describe observed responses to income, not whether a good is good or bad.

Two goods are **substitutes** when a higher price for one increases demand for the other. They are **complements** when a higher price for one reduces demand for the other.

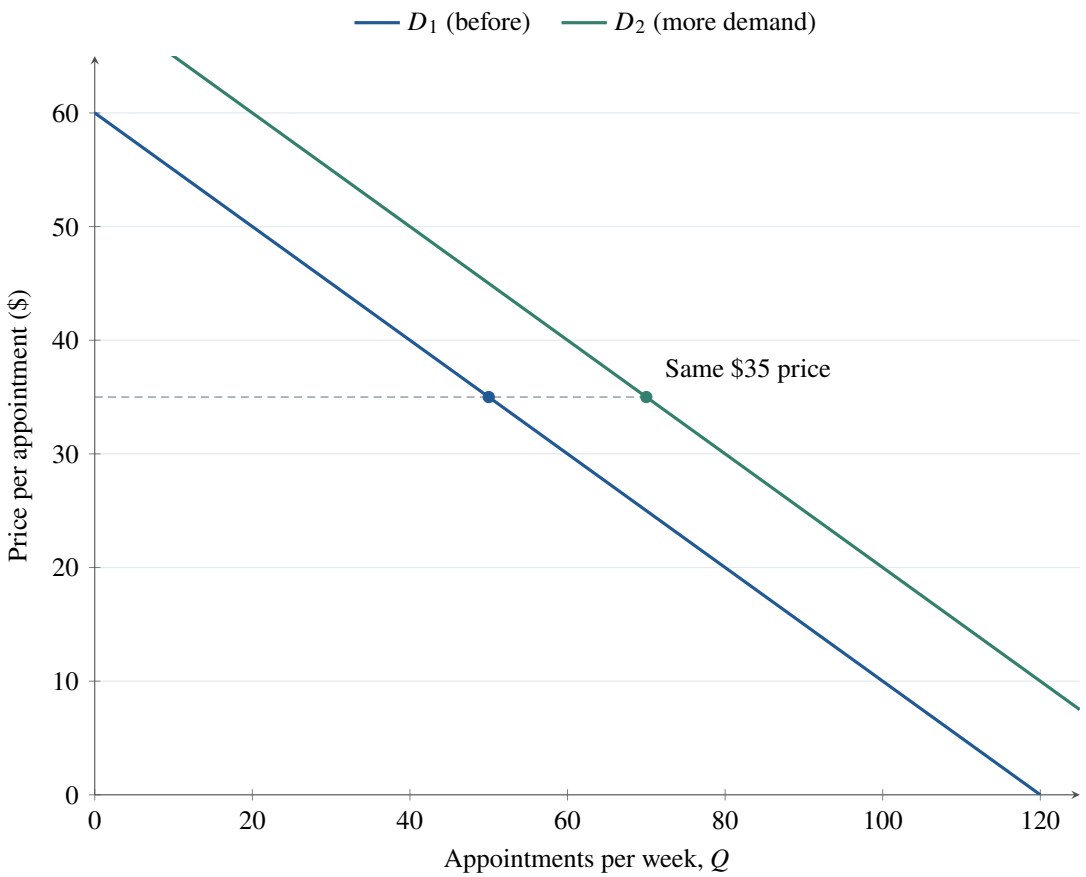


Figure 2. An increase in demand. At the *same* price of $35, customers now want 70 rather than 50 appointments. This is a rightward shift, not movement along D_1.

Read the graph. Hold the price fixed at $35 and trace horizontally. The first curve gives 50 appointments, the second 70. The curve changes position because something *other than the market price* changed. Later we will add supply to find how the *equilibrium* price also responds.

Demand shifter	**What changes, and why?**
Number of buyers	More people start cycling: more potential repair customers at every price. Demand shifts right.
Income	If professional servicing is a normal good for these riders, higher income shifts demand right. If it is inferior for some riders, the response can go the other way.
Prices of related goods	A rise in the price of *replacement bicycles* may increase demand for repairs if repair and replacement are substitutes. A rise in the price of a complementary service may reduce demand.
Tastes	Greater interest in cycling increases the desire for repairs at a given price. Demand shifts right.
Expectations	If riders expect a future rise in repair prices and can bring work forward, current demand can rise. The effect depends on their ability to change timing.

KEY RESULT Movement along demand versus shift in demand

The good's own price changes: move along the existing demand curve. **Income, preferences, other goods' prices, expectations, or buyers change:** shift demand. To decide which, ask: *At the same price as before, would customers now want a different quantity?* If yes, demand has shifted.

2. The law of supply

Demand describes buyers. Supply describes sellers. A workshop considers the revenue from another repair against the cost of parts, labor, and time. A higher selling price makes it worthwhile to accept appointments that would not have covered costs at a lower price.

DEFINITION Quantity supplied, supply schedule, and supply curve

Quantity supplied is the amount sellers are willing and able to offer at a particular price during a stated period. A **supply schedule** lists quantities supplied at different prices. A **supply curve** graphs the price–quantity relationship. As with demand, price goes on the vertical axis.

KEY RESULT The law of supply — required principle 2

Other things equal, a higher price causes a higher quantity supplied; a lower price causes a lower quantity supplied. This is a movement *along* the supply curve. In the usual example, the curve slopes upward because more production requires using increasingly costly resources or effort.

Worked example: Read a supply schedule

Suppose workshops can collectively fill 20 appointments at $20 each but 60 at $40. At the higher price, additional evening hours and more costly repairs are worthwhile. No change in technology, spare-parts prices, or the number of workshops is being assumed.

Price ($)	20	25	30	35	40	45	50
Quantity supplied (weekly)	20	30	40	50	60	70	80

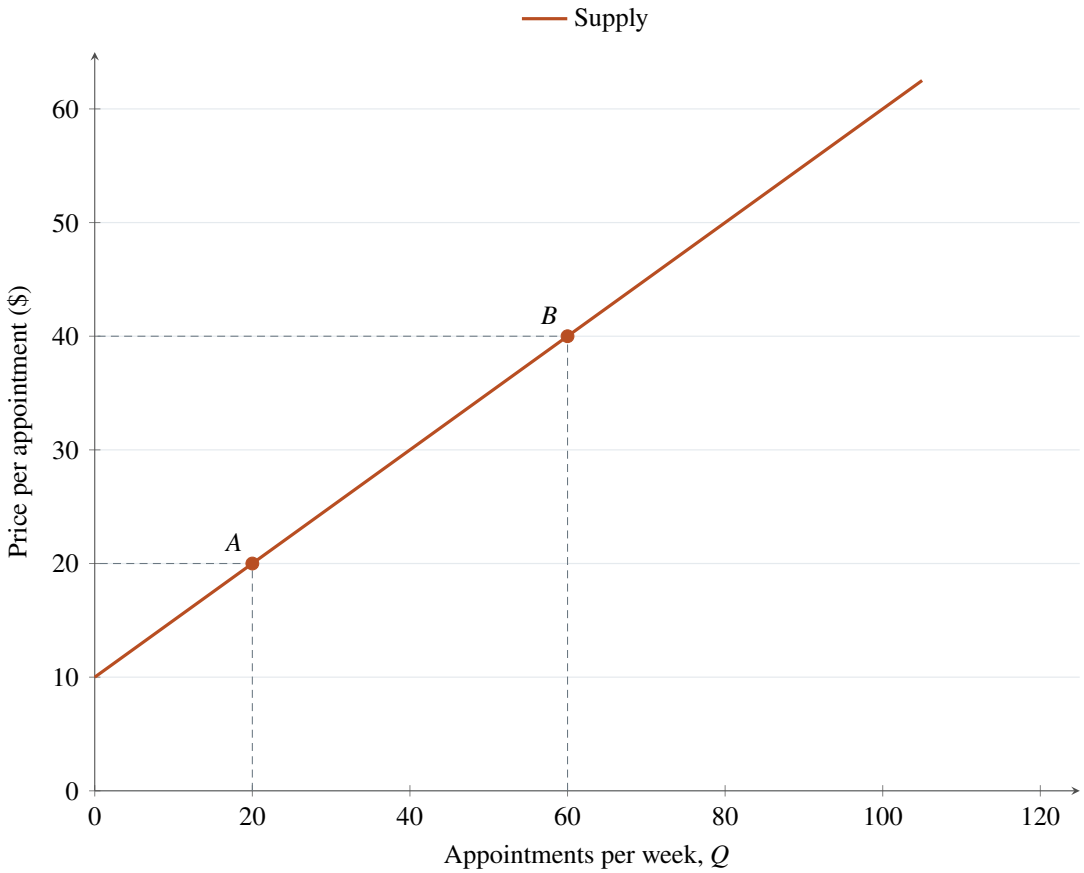


Figure 3. The law of supply. A higher selling price moves the market from *A* to *B* along an unchanged supply curve.

Read the graph. At $20 workshops offer 20 appointments. At $40 they offer 60. The price increase makes extra appointments worthwhile for sellers. The law of supply concerns *quantity supplied at alternative prices*, not the actual quantity traded when the market is out of equilibrium.

Individual supply and market supply

Market supply is calculated in exactly the same way as market demand: *add quantities at the same price*. For example, if two groups of repair shops can each supply 25 appointments at $35, total market supply is 50 appointments. A market could have many more shops; we use two groups only to make the arithmetic visible.

> **KEY RESULT Market supply is horizontal addition**
>
> At a given price, market quantity supplied is the **sum of the quantities offered by individual sellers**. As the market price rises, existing firms may expand output. New sellers may also enter over time; entry is a *shift in market supply* when the number of sellers changes.

What shifts the supply curve?

A workshop facing a higher price for brake cables must spend more to deliver each repair. It offers fewer appointments at *any given repair price*. That is a *decrease in supply*: the curve shifts left. A new tool that lets mechanics complete repairs faster has the opposite effect: supply shifts right. The important point is that the *cost or capacity changed*, not the repair price itself.

Supply shifter	Economic reasoning
Input prices	Higher wages or parts costs make repairs more costly: supply shifts left. Lower costs shift it right.
Technology	A reliable diagnostic tool reduces the time needed per job: more repairs can be offered at each price; supply shifts right.
Number of sellers	Additional independent repair shops expand market supply; closures reduce it.
Expectations	A workshop expecting much higher future prices may delay work that can be postponed, reducing current supply. For a service that cannot be stored, the timing response may be limited.

KEY RESULT Movement along supply versus shift in supply

The repair price itself changes: move along the existing supply curve. **Production costs, technology, expectations, or seller numbers change:** shift the entire supply curve. An increase in supply means a rightward shift: more offered *at the same price*, not necessarily a higher selling price.

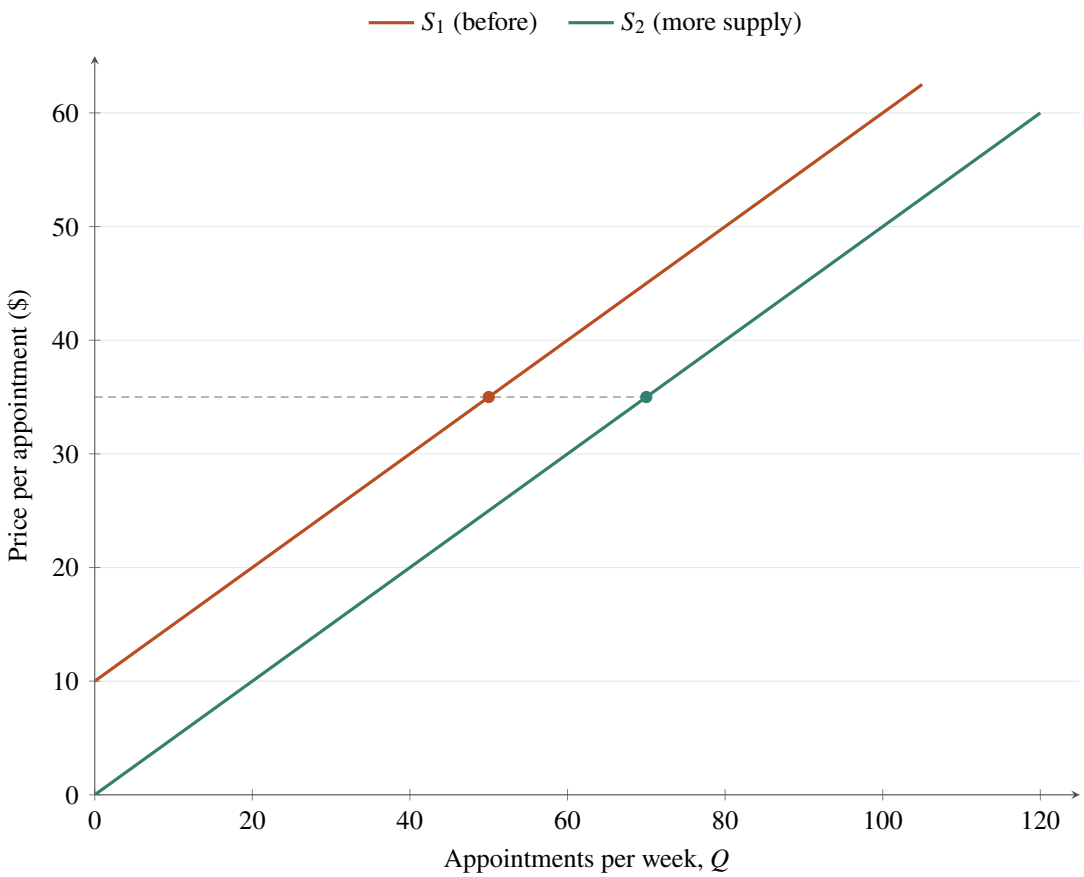


Figure 4. An increase in supply. At the same price of $35, shops can now provide 70 rather than 50 appointments. Costs have fallen or capacity has risen.

Read the graph. Keeping price at $35 isolates the change in supply: more appointments are offered regardless of the market price. The rightward shift is not caused by the repair price rising. When we add demand, this improvement in supply will usually *reduce* the equilibrium price.

3. Equilibrium

We now bring demand and supply together. At every possible price, buyers announce how many appointments they want and sellers how many they are prepared to offer. Most prices leave the two plans out of balance. The equilibrium price is the one at which they match.

> **DEFINITION Market equilibrium**
>
> **Equilibrium price** is the price at which **quantity demanded equals quantity supplied**. **Equilibrium quantity** is the number of units both sides plan to trade at that price. On a supply-and-demand graph, equilibrium is the intersection of the two curves.

Price per repair	Quantity demanded	Quantity supplied	What happens?
\$20	80	20	Shortage of 60
\$25	70	30	Shortage of 40
\$30	60	40	Shortage of 20
\$35	50	50	**Equilibrium**
\$40	40	60	Surplus of 20
\$45	30	70	Surplus of 40
\$50	20	80	Surplus of 60

Look first at the middle row. At \$35 per repair, customers want 50 appointments and workshops want to provide exactly 50. Thus the equilibrium price is \$35 and equilibrium quantity is 50 appointments per week. The table tells us *which* price clears the market; the graph shows *why* this price is the intersection.

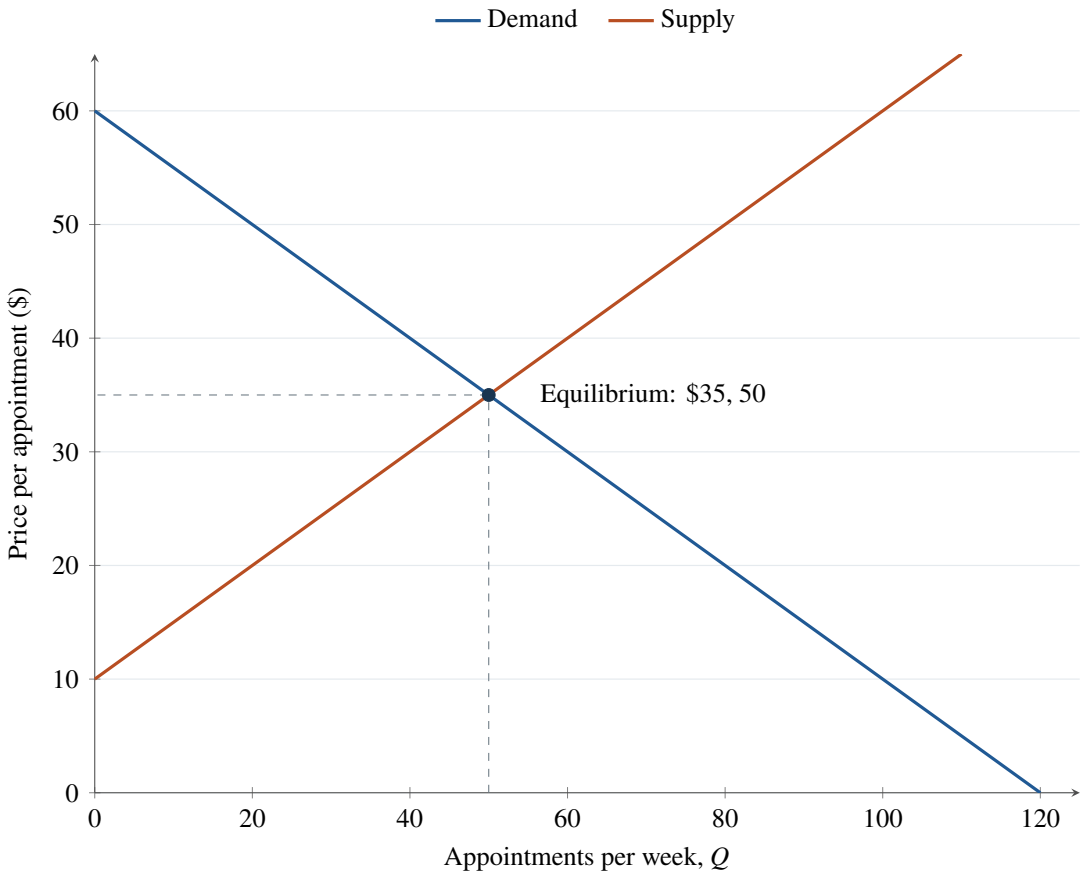


Figure 5. Supply meets demand. At \$35 per appointment, 50 appointments are both demanded and supplied. No buyer or seller needs to change plans solely because of a shortage or surplus.

Read the graph. Locate the crossing of the blue demand curve and the orange supply curve. Read across to find the price and down to find the quantity. It is a single point, not the whole demand or supply curve.

DEFINITION Shortage and surplus

A **shortage** (excess demand) occurs at a given price when quantity demanded is **greater** than quantity supplied. Its size is $Q_D - Q_S$.
A **surplus** (excess supply) occurs when quantity supplied is **greater** than quantity demanded. Its size is $Q_S - Q_D$. These are differences between *planned* quantities at the same price.

Why does a shortage push price upward?

At a price of $25, buyers seek 70 repairs but shops offer only 30. There is a **shortage of 40 appointments**, not 40 completed repairs. Some customers cannot obtain appointments at that price. If prices can adjust, workshops have an incentive to charge more; a higher price makes additional slots worth offering and prompts some buyers to postpone less urgent repairs. The shortage narrows as Q_S rises and Q_D falls.

Why does a surplus push price downward?

At $45, shops offer 70 slots but customers want only 30. There is a **surplus of 40 appointments**: unused capacity at the quoted price. To attract customers, shops have an incentive to cut prices. Lower prices encourage more bookings and make fewer costly slots worth offering. The surplus narrows as Q_D rises and Q_S falls.

KEY RESULT The law of supply and demand — required equilibrium mechanism

With flexible prices and the usual downward-sloping demand and upward-sloping supply curves: **a shortage creates upward pressure on price, while a surplus creates downward pressure on price**. These forces tend to move the market toward the price at which $Q_D = Q_S$. Prices do not necessarily adjust immediately in practice.

Common mistake. A shortage does *not* mean demand must have shifted right. It can arise simply because a quoted price is *below* equilibrium. Similarly, a surplus at a price above equilibrium does not prove that supply increased. Always identify whether a *curve moved* or we are looking at a different point on the same curves.

Why does equilibrium matter beyond the graph?

Price tells buyers how scarce an appointment is and tells sellers how worthwhile it is to provide another one. When a repair shop becomes more expensive to operate, its supply changes, and prices transmit that information to customers without every buyer having to know the workshop's costs. Equilibrium coordinates their plans. This does not, by itself, prove that every outcome is fair or socially efficient; those questions require additional assumptions and the welfare analysis of Chapter 7.

4. Comparative statics

Equilibrium changes when the conditions affecting demand or supply change. Economists call the comparison of an initial equilibrium with a new one **comparative statics**. The aim is to predict the *direction* of the change in price and quantity, not to guess a number unsupported by the information given.

DEFINITION Comparative statics

Comparative statics compares two equilibria: one before and one after a change in an underlying condition. It answers questions such as: *Does price rise or fall? Does quantity rise or fall?* It is not a theory of how many days the adjustment takes.

KEY RESULT Three-step method — required principle 4

1. Identify the market and the event. Does the event change buyers' willingness to purchase (D), sellers' willingness to provide (S), or both?

2. Decide the direction. At the *old price*, does the event increase or decrease quantity demanded or supplied? Move the affected curve right or left.

3. Compare the two intersections. Read the new equilibrium *price and quantity*. Explain the shortage or surplus that initially appears at the old price and drives adjustment.

An increase in demand: price and quantity both rise

Worked example: More commuters begin cycling

Suppose additional commuters begin riding bicycles. At every repair price, more people now want appointments: **demand shifts right**. Supply itself has not changed. At the old price of $35, buyers now want 70 appointments, but sellers offer only 50. The *new shortage* puts upward pressure on price. As the price rises, sellers supply more and buyers reduce the number of appointments they demand along the new demand curve. In our illustration the new equilibrium is **$40 and 60 appointments**.

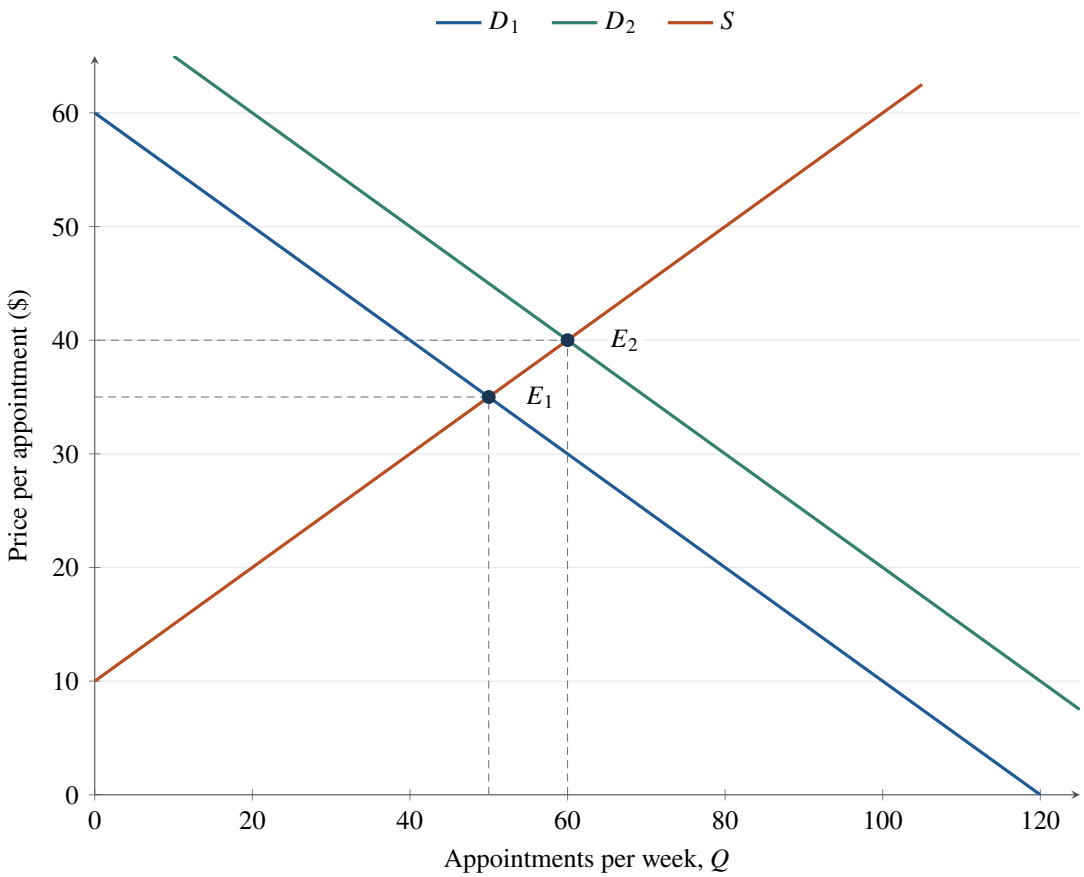


Figure 6. Demand increases. The equilibrium moves from E_1 (\$35, 50) to E_2 (\$40, 60). A rightward demand shift raises both equilibrium price and quantity.

Read the graph. Compare only the dark dots marking the two intersections. The new dot is *higher and to the right*. Higher demand also causes a *movement along* the unchanged supply curve as sellers respond to the higher price. It does *not* itself shift supply.

An increase in supply: price falls and quantity rises

Worked example: A faster repair tool cuts workshop costs

Suppose shops adopt equipment that reduces time per repair. They offer 70 appointments rather than 50 at the old \$35 price: **supply shifts right**. Buyers initially want only 50 appointments at that price, so the market develops a surplus. Shops lower prices to fill the extra capacity. The lower price encourages customers to book more repairs along the *unchanged demand curve*. In our illustration the new equilibrium is **\$30 and 60 appointments**.

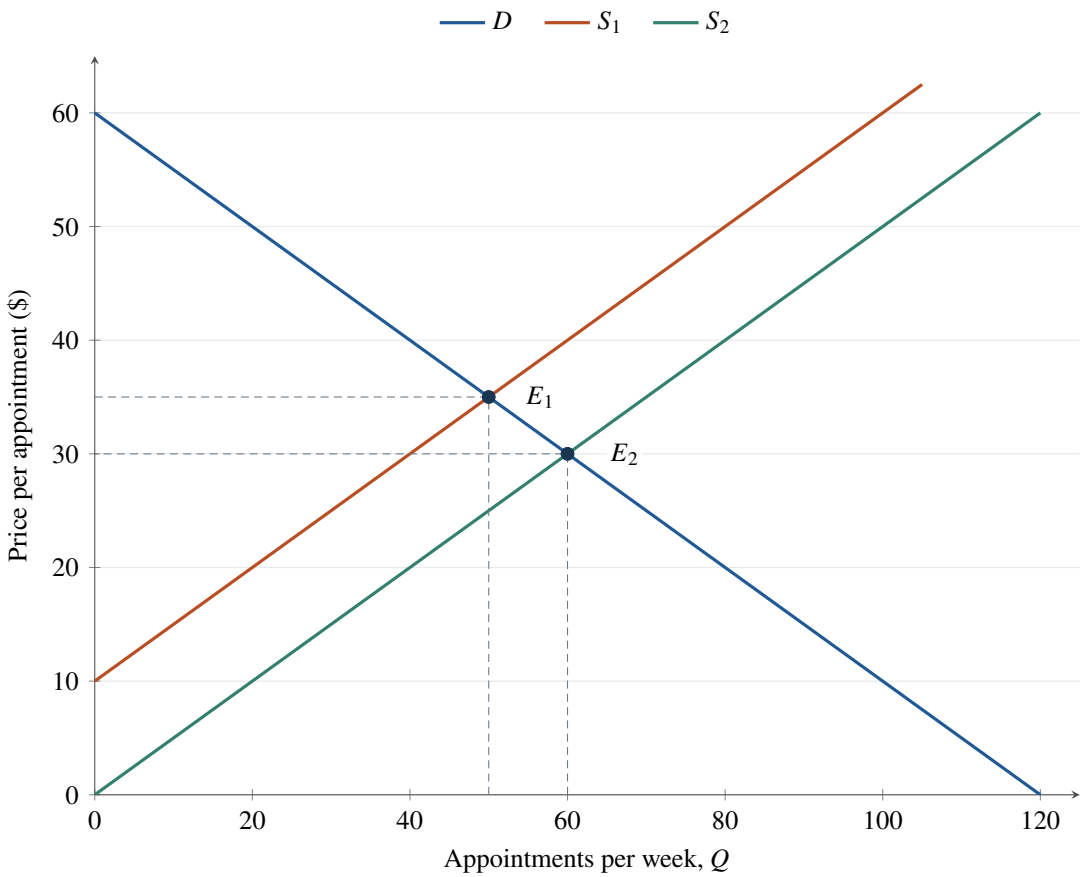


Figure 7. Supply increases. The equilibrium moves from E_1 ($35, 50) to E_2 ($30, 60). More supply lowers price while expanding the quantity traded.

Read the graph. The new equilibrium is *lower and to the right*. The fall in price raises quantity demanded *along* the fixed demand curve. It would be incorrect to say that the fall in price shifts demand right.

Decreases in demand or supply

The reverse changes are equally important. If fewer riders want repairs at every price, demand shifts left. Price *and* quantity both fall. If workshops face higher parts costs, supply shifts left. Price rises, but quantity falls. These are predictions about the new *equilibrium*, not statements that prices always rise when supply shrinks irrespective of what else changes.

Only one curve moves	At the old price	New equilibrium
Demand increases (right)	Shortage: more buyers want the good	Price ↑, quantity ↑
Demand decreases (left)	Surplus: fewer buyers want the good	Price ↓, quantity ↓
Supply increases (right)	Surplus: more sellers offer the good	Price ↓, quantity ↑
Supply decreases (left)	Shortage: fewer sellers offer the good	Price ↑, quantity ↓

When both curves shift, one answer may be uncertain

Consider two events occurring together: new riders raise demand *and* better equipment expands supply. Both changes tend to increase quantity traded, so equilibrium **quantity definitely rises** in the usual model. But demand pushes the price upward while supply pushes it downward. Without knowing the relative sizes of the shifts, we **cannot determine the direction of price**. Stating *ambiguous* is the correct economic conclusion, not a failure to finish the problem.

Worked example: Two possible outcomes from the same kinds of shocks

Initially price is \$35 and quantity is 50. Suppose demand increases by 30 appointments at each price while supply increases by only 10. The new equilibrium is **\$40 and 70 appointments**: the demand increase dominates in its effect on price. Reverse the sizes: demand increases by 10 and supply by 30. The new equilibrium becomes **\$30 and 70 appointments**. Quantity rises in both cases; price moves in opposite directions. The exact figures are invented to illustrate why the relative sizes matter.

KEY RESULT All nine directions of comparative statics

Read across the row for the change in demand and down the column for the change in supply. Arrows show changes in *equilibrium* price P and quantity Q. **Ambiguous** means that the direction depends on the relative sizes of the shifts.

	Supply decreases	**No supply change**	**Supply increases**
Demand decreases	P ?, $Q \downarrow$	$P \downarrow, Q \downarrow$	$P \downarrow, Q$?
No demand change	$P \uparrow, Q \downarrow$	No change	$P \downarrow, Q \uparrow$
Demand increases	$P \uparrow, Q$?	$P \uparrow, Q \uparrow$	P ?, $Q \uparrow$

Changes in one market can affect other markets

Suppose manufacturers face a shortage of bicycle tires. A higher price of tires raises repair shops' input costs and may shift the *supply of repair appointments* left. Independently, if cyclists respond to more expensive cycling by switching to transit, demand for transit passes may rise. These are two different markets. In each, identify the good on the axes before deciding what shifts.

KEY RESULT An application checklist for unfamiliar examination questions

1. Name the market. Exactly which good's price and quantity are on the axes?
2. Identify the shock. Is it a change in the good's own price, a buyer-side factor, or a seller-side factor?
3. Determine movement or shift. A change in the good's own price moves along a fixed curve; a change in another determinant shifts a curve.
4. Draw the new intersection. State what happens to equilibrium P and Q. If both curves shift, say explicitly when one effect is ambiguous.
5. Explain why. Identify any temporary shortage or surplus at the old price that gives buyers or sellers an incentive to adjust.